STATIC ANALYSIS & CODE QUALITY REPORT

Watson (td-watson)

Student	Anas Imran
Course	Software Quality Engineering (SQE)
Branch	static-analysis-assignment
Baseline	dfba9af — Baseline imported from original project
Final improvement	17f76db — Apply static analysis quality improvements
GitHub	github.com/AnasImran-146/Watson
Upstream	github.com/jazzband/Watson
Date	27 September 2026

Purpose: establish a baseline, interpret static-analysis findings as software-quality concerns, perform human review, apply controlled improvements, and verify behavior with the existing test suite.

1. Project Description and Code Statistics

Watson (td-watson) is an open-source command-line application for tracking time spent on projects and generating reports. The selected project is the Python source package under `watson/` from the Jazzband Watson repository. The original project was preserved as a baseline and the assignment work was performed on a separate branch.

The analyzed source package contains **10 Python source files** under `watson/`. The largest concentration of Pylint findings is in `watson/cli.py`, followed by `watson/watson.py` and `watson/frames.py`.

- Student fork: https://github.com/AnasImran-146/Watson
- Assignment branch: static-analysis-assignment
- Original upstream: https://github.com/jazzband/Watson
- Baseline commit: dfba9af — Baseline imported from original project
- Improvement commit: 17f76db — Apply static analysis quality improvements
- Complete Pylint artifacts: baseline_pylint_report.txt, configured_pylint_report.txt, final_pylint_report.txt
- Git history evidence: git_history.txt

2. Baseline and Original Source

The original Watson source was cloned from the upstream Jazzband repository and preserved before quality changes. The baseline commit is `dfba9af`.

Baseline tests: 346 passed in 1.63 seconds.

Baseline Pylint: Pylint 4.0.8 on Python 3.12.3; score 7.56/10 with 256 findings. The complete baseline output is reproduced in Appendix A and saved as `baseline_pylint_report.txt`.

3. Initial Pylint Results

Metric	Baseline
Pylint score	7.56 / 10
Total findings	256
Convention (C)	129
Warning (W)	71
Refactor (R)	53
Error (E)	3
Fatal (F)	0

Most frequent messages: C0209 75; C0116 24; W0621 20; W0707 15; W0622 13; R0917 11; R0913 11; C0114 9; W0613 8; R1705 8.

File	Findings	Share
watson/cli.py	121	47.3%
watson/watson.py	55	21.5%
watson/frames.py	30	11.7%
watson/utils.py	24	9.4%
watson/autocompletion.py	9	3.5%

File	Findings	Share
watson/config.py	8	3.1%
watson/__main__.py	3	1.2%
watson/fullmoon.py	3	1.2%
watson/version.py	2	0.8%
watson/__init__.py	1	0.4%

A key observation is that 75 of the 256 findings are the same C0209 style message. Therefore, the raw count does not represent 256 independent quality defects.

4. Classification of Findings

Quality category	Examples	Typical effect
Maintainability / complexity	R0912, R0914, R0915, R0913	Harder to understand, test and modify
Architecture / coupling	R0401	More dependency complexity
Error handling / reliability	W0707, W1514, W3101	Reduced diagnostic quality or robustness
Naming / readability	W0622, W0621, C0103	Less clear intent/scope
Documentation	C0114, C0115, C0116	Harder maintenance/onboarding
Style / consistency	C0209, C0411, R1705	Consistency/readability
API/interface design	R0913, R0917, R0902	Interfaces carrying too much responsibility
Legacy/modernization	R0205, R1725	Older Python style

5. Ten Detailed Pylint Findings

1. Cyclic import

Location: `watson/utils.py:1` **ID:** `R0401` **Quality type:** Architecture / coupling

Analysis: Cyclic dependencies involve watson, watson.watson, watson.utils and watson.frames. This increases coupling and makes module changes harder to reason about.

Classification: A — should fix

Action: Identified as a structural improvement area; not fully refactored because of regression risk and the limited assignment scope.

2. Too many branches in edit

Location: `watson/cli.py:1273` **ID:** `R0912` **Quality type:** Complexity / maintainability

Analysis: 16 branches versus the default threshold of 12. The function combines selection and validation paths.

Classification: A — should fix

Action: Documented for refactoring; a large CLI rewrite was avoided to protect behavior.

3. Too many statements in edit

Location: `watson/cli.py:1273` **ID:** `R0915` **Quality type:** Complexity / maintainability

Analysis: 54 statements versus the default threshold of 50. The function is long and procedural.

Classification: A — should fix

Action: Candidate for extracting frame selection, editor preparation, validation and update helpers.

4. Too many local variables in report

Location: `watson/watson.py:460` **ID:** `R0914` **Quality type:** Cohesion / complexity

Analysis: 27 locals versus the default threshold of 15. This suggests several operations are coordinated in one function.

Classification: A — should fix

Action: Candidate for decomposition into report-generation helpers.

5. Too many arguments in report

Location: `watson/watson.py:460` **ID:** `R0913` **Quality type:** Interface design

Analysis: 14 arguments exceed both the default threshold of 5 and configured threshold of 8.

Classification: B — context-dependent

Action: Reporting filters can legitimately create larger APIs; retained while keeping the warning visible.

6. Missing exception chaining

Location: `watson/watson.py:85` **ID:** `W0707` **Quality type:** Error handling / diagnosability

Analysis: ValueError is converted to WatsonError without explicit raise-from chaining, reducing causal context.

Classification: A — should fix

Action: Identified but not implemented; an earlier edit attempt was reverted after indentation risk.

7. Built-in id shadowed

Location: `watson/frames.py:11` **ID:** `W0622` **Quality type:** Naming / readability

Analysis: The Frame constructor used id as a parameter, shadowing Python's built-in id().

Classification: A — should fix

Action: Implemented: renamed to frame_id and related lookup variable; 346 tests still passed.

8. Missing function docstring

Location: `watson/watson.py:168` **ID:** `C0116` **Quality type:** Documentation

Analysis: A method has no function/method docstring.

Classification: B — context-dependent

Action: Not changed in the limited refactoring scope.

9. Repeated f-string recommendation

Location: `Multiple files` **ID:** `C0209` **Quality type:** Style / consistency

Analysis: 75 occurrences of .format() where Pylint recommends f-strings.

Classification: C — acceptable in current scope

Action: Not mass-refactored because it would mainly optimize the score rather than address higher-value concerns.

10. Useless object inheritance

Location: `watson/frames.py:63` **ID:** `R0205` **Quality type:** Modernization / readability

Analysis: Span and Frames explicitly inherit from object under Python 3.

Classification: C — low priority

Action: Harmless modernization; not materially important to current behavior or architecture.

6. Five Manual Code-Quality Issues

6.1 HTTP status validation uses assert

The HTTP methods in watson/watson.py use assertions for response status. Assertions can be disabled under Python optimization, so explicit status checks are more reliable. **Impact:** Reliability and error handling. **Suggested improvement:** Replace assertions with explicit checks and Watson-specific exceptions.

6.2 parse_tags() is overly compressed

watson/utils.py combines filter, map, methodcaller, takewhile and a generator expression in one dense expression. **Impact:** Readability and maintainability. **Suggested improvement:** Split into named steps or a helper.

6.3 Broad except Exception in safe_save()

The save path catches Exception broadly before cleanup/re-raising. **Impact:** Error-handling precision. **Suggested improvement:** Catch expected filesystem/serialization exceptions where possible.

6.4 edit() has several responsibilities

The function handles frame selection, editor preparation, JSON parsing, project/tag handling, date parsing, validation, update and persistence. **Impact:** Cohesion, testability and maintainability. **Suggested improvement:** Extract logically independent operations.

6.5 Repeated HTTP request/error handling

Several HTTP methods repeat request execution, status validation and error conversion. **Impact:** Duplication and maintenance cost. **Suggested improvement:** Centralize common request/error handling while preserving endpoint-specific behavior.

7. Pylint Configuration Decisions

Setting	Default	Configured	Rationale
max-args	5	8	Some Watson CLI/report APIs legitimately accept more parameters; extreme cases remain visible.
max-locals	15	20	Some reporting/CLI functions naturally carry more intermediate values; check remains active.
max-statements	50	60	Allows a small amount of project-specific procedural code while detecting substantially larger functions

The configuration was conservative. No broad message categories were disabled simply to inflate the score. The configured run moved the score from 7.56 to 7.64.

8. Improvements and Before/After Evidence

Rename id to frame_id

Before: Frame.__new__(..., id, ...) and related lookup code used the built-in name id.
After: Parameter and related variable renamed to frame_id.
Quality effect: Clearer intent and no built-in shadowing; all 346 tests passed.

Establish explicit baseline commit

Before: No assignment-specific baseline marker.
After: dfba9af records the imported original state.
Quality effect: Reproducibility and traceability.

Add project-specific Pylint configuration

Before: Pylint ran with defaults.
After: .pylintrc records three justified threshold changes.
Quality effect: Analysis better reflects the project while keeping useful checks.

Keep environment files out of version control

Before: Assignment environment needed protection from accidental virtual-environment commits.
After: .gitignore includes .venv/.
Quality effect: Cleaner repository history and artifacts.

Preserve complete analysis artifacts

Before: Results existed as terminal output.
After: Baseline/configured/final reports and Git history are saved.
Quality effect: Reproducibility and auditability.

Scope note: The assignment did not perform a large-scale refactoring of cli.py or watson.py. Several higher-impact recommendations remain documented rather than being claimed as completed fixes.

9. Testing and Verification

Stage	Result
Baseline	346 passed in 1.63 s
Final	346 passed in 2.29 s

No regression was detected by the existing automated suite. The final branch also contains the Pylint reports and assignment-specific commits.

10. Final Pylint Results and Comparison

Metric	Baseline	Configured	Final	Baseline → Final
Score	7.56	7.64	7.66	+0.10
Total findings	256	254	252	-4

The configured run reduced threshold-related findings after the three settings were adjusted. The final source change produced a further small reduction. The final report still contains significant findings, including cyclic imports, complexity warnings, missing exception chaining and repeated style/documentation messages. Therefore, the numerical score should be interpreted alongside the detailed findings and human review.

11. Automated Analysis vs Human Review

Pylint systematically detects repeated patterns such as naming problems, missing documentation, complexity thresholds, import problems, exception-handling patterns and style inconsistencies. It also provides a reproducible numerical baseline.

Human review adds context that a rule-based analyzer does not fully capture. The use of assert for HTTP response validation, the dense parse_tags() expression and the multiple responsibilities of edit() require understanding behavior and design rather than simply counting syntax patterns.

The combined approach is stronger than either method alone: static analysis provides breadth and repeatability, while human review provides context, architectural reasoning and prioritization.

12. Git Commit History and Repository Progression

Commit	Purpose
b820093	Bump release to 2.1.0 — upstream project history
df4b720	Use the mock standard module — upstream project history
d9de4fc	Fix "start --at" fail when stopping a current project (#480) — upstream baseline history
dfba9af	Baseline imported from original project — assignment baseline
17f76db	Apply static analysis quality improvements — assignment changes
Subsequent assignment commit(s)	Save Pylint reports and supporting evidence such as git_history.txt and python_file_count.txt

GitHub repository: https://github.com/AnasImran-146/Watson

Assignment branch: static-analysis-assignment

The complete Git history is also exported to git_history.txt in the repository.

13. Conclusion

The exercise established a reproducible baseline for Watson, categorized 256 initial Pylint findings into meaningful quality themes, investigated ten representative findings, identified five additional issues through manual review, configured Pylint using project-specific thresholds, made a small regression-safe source improvement, and verified the full test suite.

The Pylint score moved from **7.56/10 to 7.66/10**, while the test suite remained at **346 passing tests**. The small score change is informative: the project contains both low-impact style findings and deeper maintainability/design concerns. Static analysis is therefore most useful when its findings are interpreted in context rather than treated as a single quality score.

Appendix A — Complete Initial Pylint Output

The following is the complete baseline output captured from the initial Pylint run. The same output is committed as baseline_pylint_report.txt in the repository.

```
************* Module watson
watson/__init__.py:1:0: C0114: Missing module docstring (missing-module-docstring)
************* Module watson.frames
watson/frames.py:1:0: C0114: Missing module docstring (missing-module-docstring)
watson/frames.py:10:0: C0115: Missing class docstring (missing-class-docstring)
watson/frames.py:11:4: R0913: Too many arguments (6/5) (too-many-arguments)
watson/frames.py:11:4: R0917: Too many positional arguments (6/5) (too-many-positional-arguments)
watson/frames.py:11:43: W0622: Redefining built-in 'id' (redefined-builtin)
watson/frames.py:24:12: C0415: Import outside toplevel (watson.WatsonError) (import-outside-toplevel)
watson/frames.py:25:12: W0707: Consider explicitly re-raising using 'raise WatsonError('Error converting date: {}'.format(e)) from e' (raise-missing-from)
watson/frames.py:25:30: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/frames.py:39:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/frames.py:47:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/frames.py:63:0: C0115: Missing class docstring (missing-class-docstring)
watson/frames.py:63:0: R0205: Class 'Span' inherits from object, can be safely removed from bases in Python3 (useless-object-inheritance)
watson/frames.py:69:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/frames.py:76:0: C0115: Missing class docstring (missing-class-docstring)
watson/frames.py:76:0: R0205: Class 'Frames' inherits from object, can be safely removed from bases in Python3 (useless-object-inheritance)
watson/frames.py:90:8: R1705: Unnecessary "elif" after "return", remove the leading "el" from "elif" (no-else-return)
watson/frames.py:122:31: W0622: Redefining built-in 'id' (redefined-builtin)
watson/frames.py:128:12: W0707: Consider explicitly re-raising using 'except StopIteration as exc' and 'raise KeyError('Frame with id {} not found.'.format(id
watson/frames.py:128:27: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/frames.py:135:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/frames.py:141:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/frames.py:141:4: R0913: Too many arguments (6/5) (too-many-arguments)
watson/frames.py:141:4: R0917: Too many positional arguments (6/5) (too-many-positional-arguments)
watson/frames.py:141:57: W0622: Redefining built-in 'id' (redefined-builtin)
watson/frames.py:148:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/frames.py:151:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/frames.py:151:4: R0913: Too many arguments (6/5) (too-many-arguments)
watson/frames.py:151:4: R0917: Too many positional arguments (6/5) (too-many-positional-arguments)
watson/frames.py:186:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/frames.py:5:0: C0411: standard import "collections.namedtuple" should be placed before third party import "arrow" (wrong-import-order)
************* Module watson.fullmoon
watson/fullmoon.py:1:0: C0114: Missing module docstring (missing-module-docstring)
watson/fullmoon.py:229:12: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/fullmoon.py:2:0: C0411: standard import "bisect" should be placed before third party import "arrow" (wrong-import-order)
************* Module watson.autocompletion
watson/autocompletion.py:1:0: C0114: Missing module docstring (missing-module-docstring)
watson/autocompletion.py:46:18: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/autocompletion.py:53:4: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no-else-return)
watson/autocompletion.py:62:22: W0613: Unused argument 'args' (unused-argument)
watson/autocompletion.py:83:4: R1705: Unnecessary "elif" after "return", remove the leading "el" from "elif" (no-else-return)
watson/autocompletion.py:91:21: W0613: Unused argument 'ctx' (unused-argument)
watson/autocompletion.py:91:26: W0613: Unused argument 'args' (unused-argument)
watson/autocompletion.py:98:18: W0613: Unused argument 'args' (unused-argument)
watson/autocompletion.py:106:20: W0613: Unused argument 'args' (unused-argument)
************* Module watson.__main__
watson/__main__.py:1:0: C0114: Missing module docstring (missing-module-docstring)
watson/__main__.py:3:0: E1120: No value for argument 'ctx' in function call (no-value-for-parameter)
watson/__main__.py:3:0: E1120: No value for argument 'color' in function call (no-value-for-parameter)
************* Module watson.watson
watson/watson.py:1:0: C0114: Missing module docstring (missing-module-docstring)
watson/watson.py:17:0: C0115: Missing class docstring (missing-class-docstring)
watson/watson.py:21:0: C0115: Missing class docstring (missing-class-docstring)
watson/watson.py:25:0: C0115: Missing class docstring (missing-class-docstring)
watson/watson.py:25:0: R0205: Class 'Watson' inherits from object, can be safely removed from bases in Python3 (useless-object-inheritance)
watson/watson.py:25:0: R0902: Too many instance attributes (15/7) (too-many-instance-attributes)
watson/watson.py:67:40: W0622: Redefining built-in 'type' (redefined-builtin)
watson/watson.py:74:17: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
watson/watson.py:82:12: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no-else-return)
watson/watson.py:85:16: W0707: Consider explicitly re-raising using 'raise WatsonError('Invalid JSON file {}: {}'.format(filename, e)) from e' (raise-missing-
watson/watson.py:86:20: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/watson.py:89:12: W0707: Consider explicitly re-raising using 'raise WatsonError('Unexpected error while loading JSON file {}: {}'.format(filename, e))
watson/watson.py:90:16: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/watson.py:116:16: W0707: Consider explicitly re-raising using 'raise ConfigurationError('Cannot parse config file: {}'.format(e)) from e' (raise-missin
watson/watson.py:117:20: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/watson.py:163:12: W0707: Consider explicitly re-raising using 'raise WatsonError('Impossible to write {}: {}'.format(e.filename, e)) from e' (raise-mis
watson/watson.py:164:16: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/watson.py:168:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/watson.py:179:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/watson.py:213:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/watson.py:233:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/watson.py:236:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/watson.py:248:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/watson.py:252:16: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/watson.py:281:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/watson.py:306:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/watson.py:335:19: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/watson.py:347:29: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/watson.py:354:8: C0415: Import outside toplevel (requests) (import-outside-toplevel)
watson/watson.py:359:27: W3101: Missing timeout argument for method 'requests.get' can cause your program to hang indefinitely (missing-timeout)
watson/watson.py:364:16: W0707: Consider explicitly re-raising using 'except Exception as exc' and 'raise WatsonError('Unable to reach the server.') from exc'
watson/watson.py:366:16: W0707: Consider explicitly re-raising using 'except AssertionError as exc' and 'raise WatsonError('An error occurred with the remote
watson/watson.py:367:20: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/watson.py:373:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/watson.py:374:8: C0415: Import outside toplevel (requests) (import-outside-toplevel)
watson/watson.py:378:23: W3101: Missing timeout argument for method 'requests.get' can cause your program to hang indefinitely (missing-timeout)
watson/watson.py:383:12: W0707: Consider explicitly re-raising using 'except Exception as exc' and 'raise WatsonError('Unable to reach the server.') from exc'
watson/watson.py:385:12: W0707: Consider explicitly re-raising using 'except AssertionError as exc' and 'raise WatsonError('An error occurred with the remote
watson/watson.py:386:16: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/watson.py:403:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/watson.py:404:8: C0415: Import outside toplevel (requests) (import-outside-toplevel)
watson/watson.py:420:23: W3101: Missing timeout argument for method 'requests.post' can cause your program to hang indefinitely (missing-timeout)
watson/watson.py:423:12: W0707: Consider explicitly re-raising using 'except Exception as exc' and 'raise WatsonError('Unable to reach the server.') from exc'
watson/watson.py:425:12: W0707: Consider explicitly re-raising using 'except AssertionError as exc' and 'raise WatsonError('An error occurred with the remote
```

```
watson/watson.py:426:16: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/watson.py:435:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/watson.py:460:4: C0116: Missing function or method docstring (missing-function-docstring)
watson/watson.py:460:4: R0913: Too many arguments (14/5) (too-many-arguments)
watson/watson.py:460:4: R0917: Too many positional arguments (14/5) (too-many-positional-arguments)
watson/watson.py:460:4: R0914: Too many local variables (27/15) (too-many-locals)
watson/watson.py:462:59: W0622: Redefining built-in 'all' (redefined-builtin)
watson/watson.py:554:30: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/watson.py:571:30: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/watson.py:362:16: W0201: Attribute '_remote_projects' defined outside __init__ (attribute-defined-outside-init)
watson/watson.py:14:0: W0611: Unused version imported from version as __version__ (unused-import)
************* Module watson.config
watson/config.py:22:4: W0221: Number of parameters was 7 in 'RawConfigParser.getint' and is now 4 in overriding 'ConfigParser.getint' method (arguments-differ
watson/config.py:22:4: W0221: Variadics removed in overriding 'ConfigParser.getint' method (arguments-differ)
watson/config.py:34:4: W0221: Number of parameters was 7 in 'RawConfigParser.getfloat' and is now 4 in overriding 'ConfigParser.getfloat' method (arguments-di
watson/config.py:34:4: W0221: Variadics removed in overriding 'ConfigParser.getfloat' method (arguments-differ)
watson/config.py:46:4: W0221: Number of parameters was 7 in 'RawConfigParser.getboolean' and is now 4 in overriding 'ConfigParser.getboolean' method (argument
watson/config.py:46:4: W0221: Variadics removed in overriding 'ConfigParser.getboolean' method (arguments-differ)
watson/config.py:91:8: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no-else-return)
watson/config.py:97:4: W0222: Signature differs from overridden 'set' method (signature-differs)
************* Module watson.version
watson/version.py:1:0: C0114: Missing module docstring (missing-module-docstring)
watson/version.py:1:0: C0103: Constant name "version" doesn't conform to UPPER_CASE naming style (invalid-name)
************* Module watson.cli
watson/cli.py:300:0: C0301: Line too long (105/100) (line-too-long)
watson/cli.py:1007:0: C0301: Line too long (103/100) (line-too-long)
watson/cli.py:1:0: C0302: Too many lines in module (1728/1000) (too-many-lines)
watson/cli.py:1:0: C0114: Missing module docstring (missing-module-docstring)
watson/cli.py:167:0: W0622: Redefining built-in 'help' (redefined-builtin)
watson/cli.py:43:0: C0115: Missing class docstring (missing-class-docstring)
watson/cli.py:46:8: R1725: Consider using Python 3 style super() without arguments (super-with-arguments)
watson/cli.py:54:15: R1725: Consider using Python 3 style super() without arguments (super-with-arguments)
watson/cli.py:65:16: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:67:21: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:85:0: C0115: Missing class docstring (missing-class-docstring)
watson/cli.py:93:20: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:88:4: R1710: Either all return statements in a function should return an expression, or none of them should. (inconsistent-return-statements)
watson/cli.py:129:0: C0103: Constant name "DateTime" doesn't conform to UPPER_CASE naming style (invalid-name)
watson/cli.py:132:0: C0116: Missing function or method docstring (missing-function-docstring)
watson/cli.py:157:20: R1719: The if expression can be replaced with 'bool(test)' (simplifiable-if-expression)
watson/cli.py:178:35: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:183:0: R0913: Too many arguments (6/5) (too-many-arguments)
watson/cli.py:183:0: R0917: Too many positional arguments (6/5) (too-many-positional-arguments)
watson/cli.py:183:28: W0621: Redefining name 'tags' from outer scope (line 1163) (redefined-outer-name)
watson/cli.py:183:34: W0621: Redefining name 'restart' from outer scope (line 329) (redefined-outer-name)
watson/cli.py:189:15: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:192:22: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:215:0: R0913: Too many arguments (7/5) (too-many-arguments)
watson/cli.py:215:0: R0917: Too many positional arguments (7/5) (too-many-positional-arguments)
watson/cli.py:258:4: W0621: Redefining name 'tags' from outer scope (line 1163) (redefined-outer-name)
watson/cli.py:329:0: R0913: Too many arguments (6/5) (too-many-arguments)
watson/cli.py:329:0: R0917: Too many positional arguments (6/5) (too-many-positional-arguments)
watson/cli.py:329:25: W0622: Redefining built-in 'id' (redefined-builtin)
watson/cli.py:383:39: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:403:15: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:419:28: W0621: Redefining name 'tags' from outer scope (line 1163) (redefined-outer-name)
watson/cli.py:446:19: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:452:19: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:458:19: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:465:15: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:547:0: R0913: Too many arguments (18/5) (too-many-arguments)
watson/cli.py:547:0: R0917: Too many positional arguments (18/5) (too-many-positional-arguments)
watson/cli.py:547:0: R0914: Too many local variables (26/15) (too-many-locals)
watson/cli.py:547:39: W0621: Redefining name 'projects' from outer scope (line 1143) (redefined-outer-name)
watson/cli.py:547:49: W0621: Redefining name 'tags' from outer scope (line 1163) (redefined-outer-name)
watson/cli.py:548:54: W0622: Redefining built-in 'all' (redefined-builtin)
watson/cli.py:663:4: W0621: Redefining name 'report' from outer scope (line 547) (redefined-outer-name)
watson/cli.py:669:4: R1705: Unnecessary "elif" after "return", remove the leading "el" from "elif" (no-else-return)
watson/cli.py:694:25: W0613: Unused argument 'lines' (unused-argument)
watson/cli.py:701:25: W0613: Unused argument 'lines' (unused-argument)
watson/cli.py:706:15: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:707:26: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:710:26: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:716:15: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:717:26: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:720:26: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:728:15: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:741:23: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:742:39: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:745:37: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:756:11: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:757:22: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:547:0: R0912: Too many branches (14/12) (too-many-branches)
watson/cli.py:547:0: R1710: Either all return statements in a function should return an expression, or none of them should be. (inconsistent-return-statements
watson/cli.py:801:0: R0913: Too many arguments (9/5) (too-many-arguments)
watson/cli.py:801:0: R0917: Too many positional arguments (9/5) (too-many-positional-arguments)
watson/cli.py:801:47: W0621: Redefining name 'projects' from outer scope (line 1143) (redefined-outer-name)
watson/cli.py:801:57: W0621: Redefining name 'tags' from outer scope (line 1163) (redefined-outer-name)
watson/cli.py:974:0: R0913: Too many arguments (17/5) (too-many-arguments)
watson/cli.py:974:0: R0917: Too many positional arguments (17/5) (too-many-positional-arguments)
watson/cli.py:974:0: R0914: Too many local variables (29/15) (too-many-locals)
watson/cli.py:974:45: W0621: Redefining name 'projects' from outer scope (line 1143) (redefined-outer-name)
watson/cli.py:974:55: W0621: Redefining name 'tags' from outer scope (line 1163) (redefined-outer-name)
watson/cli.py:975:51: W0622: Redefining built-in 'all' (redefined-builtin)
watson/cli.py:1104:17: W0621: Redefining name 'frames' from outer scope (line 1191) (redefined-outer-name)
watson/cli.py:1101:25: W0613: Unused argument 'lines' (unused-argument)
watson/cli.py:1104:12: R1704: Redefining argument with the local name 'day' (redefined-argument-from-local)
watson/cli.py:1117:12: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1118:35: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1124:12: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1126:41: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1130:36: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1131:35: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:974:0: R0912: Too many branches (13/12) (too-many-branches)
watson/cli.py:1221:0: R0913: Too many arguments (6/5) (too-many-arguments)
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watson/cli.py:1221:0: R0917: Too many positional arguments (6/5) (too-many-positional-arguments)
watson/cli.py:1244:4: W0621: Redefining name 'tags' from outer scope (line 1163) (redefined-outer-name)
watson/cli.py:1254:8: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1273:0: R0914: Too many local variables (17/15) (too-many-locals)
watson/cli.py:1273:55: W0622: Redefining built-in 'id' (redefined-builtin)
watson/cli.py:1318:4: W0621: Redefining name 'start' from outer scope (line 215) (redefined-outer-name)
watson/cli.py:1319:4: W0621: Redefining name 'stop' from outer scope (line 288) (redefined-outer-name)
watson/cli.py:1337:12: W0621: Redefining name 'tags' from outer scope (line 1163) (redefined-outer-name)
watson/cli.py:1290:22: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1359:23: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1377:25: R1735: Consider using '{"start": start, "project": project, "tags": tags}' instead of a call to 'dict'. (use-dict-literal)
watson/cli.py:1381:8: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1273:0: R0912: Too many branches (16/12) (too-many-branches)
watson/cli.py:1273:0: R0915: Too many statements (54/50) (too-many-statements)
watson/cli.py:1404:19: W0622: Redefining built-in 'id' (redefined-builtin)
watson/cli.py:1414:12: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1418:36: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1419:35: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1437:32: W0621: Redefining name 'edit' from outer scope (line 1273) (redefined-outer-name)
watson/cli.py:1458:17: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
watson/cli.py:1470:12: W0104: Statement seems to have no effect (pointless-statement)
watson/cli.py:1471:15: E1101: Module 'watson' has no 'ConfigurationError' member (no-member)
watson/cli.py:1484:8: W0707: Consider explicitly re-raising using 'except ValueError as exc' and 'raise click.ClickException("The key must have the format 'se
watson/cli.py:1490:39: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1494:16: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1528:15: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1531:15: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1544:0: R0914: Too many local variables (23/15) (too-many-locals)
watson/cli.py:1652:12: W0621: Redefining name 'start' from outer scope (line 215) (redefined-outer-name)
watson/cli.py:1656:12: W0621: Redefining name 'stop' from outer scope (line 288) (redefined-outer-name)
watson/cli.py:1682:30: W0622: Redefining built-in 'id' (redefined-builtin)
watson/cli.py:1682:34: W0621: Redefining name 'tags' from outer scope (line 1163) (redefined-outer-name)
watson/cli.py:1608:15: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1610:15: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1612:15: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1638:19: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1713:19: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1719:19: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:1726:12: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/cli.py:7:0: C0411: standard import "functools.reduce" should be placed before third party import "dateutil.tz" (wrong-import-order)
************* Module watson.utils
watson/utils.py:1:0: C0114: Missing module docstring (missing-module-docstring)
watson/utils.py:20:0: C0116: Missing function or method docstring (missing-function-docstring)
watson/utils.py:32:15: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/utils.py:46:18: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/utils.py:51:0: C0116: Missing function or method docstring (missing-function-docstring)
watson/utils.py:56:15: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/utils.py:60:24: W0622: Redefining built-in 'id' (redefined-builtin)
watson/utils.py:76:4: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no-else-return)
watson/utils.py:95:21: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/utils.py:100:21: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/utils.py:103:17: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/utils.py:122:8: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no-else-return)
watson/utils.py:125:29: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/utils.py:143:8: W0707: Consider explicitly re-raising using 'except IndexError as exc' and 'raise click.ClickException(style('error', 'No frame found f
watson/utils.py:144:27: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/utils.py:153:8: W0707: Consider explicitly re-raising using 'except KeyError as exc' and 'raise click.ClickException('{} {}.'.format(style('error', 'No
watson/utils.py:153:35: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/utils.py:159:0: C0116: Missing function or method docstring (missing-function-docstring)
watson/utils.py:184:25: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/utils.py:234:4: R1720: Unnecessary "else" after "raise", remove the "else" and de-indent the code inside it (no-else-raise)
watson/utils.py:402:20: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
watson/utils.py:17:0: C0411: third party import "click.exceptions.UsageError" should be placed before first party import "watson" and local import "fullmoon.g
watson/utils.py:1:0: R0401: Cyclic import (watson.frames -> watson.watson) (cyclic-import)
watson/utils.py:1:0: R0401: Cyclic import (watson -> watson.watson -> watson.utils) (cyclic-import)

-------------------------------------------------------------------
Your code has been rated at 7.56/10 (previous run: 7.56/10, +0.00)
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Appendix B — Analysis Artifacts

- .pylintrc — project-specific Pylint configuration
- baseline_pylint_report.txt — complete initial Pylint output
- configured_pylint_report.txt — Pylint output after configuration
- final_pylint_report.txt — final Pylint output
- git_history.txt — exported Git history
- python_file_count.txt — source-file count evidence

Repository: https://github.com/AnasImran-146/Watson

Branch: static-analysis-assignment